

Aleksandar Avdalović

Ph.D. Student in Computer Science and Engineering @ Louisiana State University

 [LinkedIn](#)  [Email](#)  [Google Scholar](#)  [Personal Website](#)  [GitHub](#)

Research Interests: Trustworthy machine learning for cyber-physical security and safety.

Methods: Uncertainty quantification with conformal prediction, topological and geometric deep learning, symbolic regression, formal verification.

Education

Louisiana State University

Ph.D. in Computer Science and Engineering

GPA: **3.85**

Advisor: Dr. Elias Bou-Harb

Committee: Dr. Xugui Zhou, Dr. Dong Lao

Baton Rouge, LA, USA

Aug. 2023 – Jul. 2027 (expected)

University of Primorska, FAMNIT

M.Sc. in Data Science

GPA: **3.89** (U.S. equivalent, ECE evaluation)

Thesis: *Translation of Random Forests into Loop-Free Imperative Programs for Formal Verification*

Advisor: Dr. Raul Barbosa (University of Coimbra)

Koper, Slovenia

Oct. 2020 – Aug. 2023

University of Primorska, FAMNIT

B.Sc. in Mathematics

GPA: **4.00** (U.S. equivalent, ECE evaluation)

Koper, Slovenia

Oct. 2017 – Jul. 2020

Additional Academic Study

University of Coimbra

Master's thesis research visit, Centre for Informatics and Systems (CISUC)

Coimbra, Portugal

Feb. 2022 – Jun. 2022

Eötvös Loránd University (ELTE)

ERASMUS+ Exchange Student, Applied Mathematics

Budapest, Hungary

Feb. 2021 – Jun. 2021

Publications

Conference Proceedings

1. A. N. Massaad, **Aleksandar Avdalović**, P. Andras, J. Khoury, and E. Bou-Harb. *CANDLE: Discerning Deep Learning Methods for CAN Bus Intrusion Detection via a Reproducible and Extensible Benchmarking Framework*. 29th International Symposium on Research in Attacks, Intrusions and Defenses (RAID), 2026. To appear. [\[code\]](#)
2. **Aleksandar Avdalović**, J. Khoury, and E. Bou-Harb. *GRACE-ICS: Graph-Aware Conformal Intervals for Industrial Operations Assurance*. IEEE/IFIP Network Operations and Management Symposium (NOMS), 2026, pp. 1–8. [\[doi\]](#) [\[code\]](#)
3. A. N. Massaad*, **Aleksandar Avdalović***, P. Andras, J. Khoury, and E. Bou-Harb. *Beyond Arbitrary Thresholds: Conformal Prediction for Trustworthy CAN Bus Intrusion Detection*. IEEE International Conference on Communications (ICC), 2026, pp. 1–6. [\[doi\]](#)
**Equal contribution.*
4. **Aleksandar Avdalović**, J. Khoury, A. F. Taha, and E. Bou-Harb. *Enhancing Network Security Management in Water Systems using FM-based Attack Attribution*. IEEE/IFIP Network Operations and Management Symposium (NOMS), 2025, pp. 1–10. [\[doi\]](#) [\[arXiv\]](#)

5. I. T. Leong, **Aleksandar Avdalović**, and R. Barbosa. *Program Verification for Rigorous Analysis of Decision Tree Ensembles Against Specifications*. IEEE/IFIP International Conference on Dependable Systems and Networks Workshops (DSN-W), 2025, pp. 55–62. [doi] [code]

Journal Articles

1. **Aleksandar Avdalović**, J. Khoury, I. AlJabea, M. Hajij, and E. Bou-Harb. *Beyond Flat Models: Preserving ICS Topology using Combinatorial Complexes for Hierarchy-Aware Intrusion Detection*. IEEE Transactions on Network and Service Management. **Minor revision**.

Papers Under Review

1. **Aleksandar Avdalović**, A. N. Massaad, J. Khoury, and E. Bou-Harb. *ACID: Automatic and Comprehensible Intrusion Detection for Industrial Control Systems via Symbolic Regression*. Network and Distributed System Security Symposium (NDSS), 2027.
2. **Aleksandar Avdalović**, J. Khoury, and E. Bou-Harb. *Calibrating the Alarm: Why Deep Learning ICS Anomaly Detectors Fail at Deployment*. IEEE Transactions on Network and Service Management. [code]

Teaching & Mentorship

Graduate Research Mentor
Cyber Threat Intelligence Lab

Louisiana State University
2025 – Present

- Mentor a junior Ph.D. student on experimental design and evaluation rigor, and on bringing uncertainty quantification into vehicular network security. The work led to two co-authored papers, at IEEE ICC 2026 and RAID 2026.

Guest Lecturer
CSC 2362, Introduction to Cybersecurity

Louisiana State University
Fall 2026

- Lecture on trustworthy AI for process-level intrusion detection in industrial control systems, covering how symbolic models let engineers check a detector against known physics before deployment.

Student Math Tutor
Secondary school and university students

Slovenia
2017 – 2022

- Taught calculus, linear algebra, and discrete mathematics in one on one sessions, adapting explanations to each student's background.

Courses Ready to Teach

Undergraduate. Introduction to Cybersecurity, Introduction to Machine Learning, Fundamentals of Artificial Intelligence, Deep Learning, Data Structures and Algorithms, Database Systems, Discrete Mathematics, Probability and Statistics, Linear Algebra.

Graduate. Network and Systems Security, Statistical Machine Learning, Applied Deep Learning, Advanced Algorithms, Formal Verification, Uncertainty Quantification, Geometric and Topological Deep Learning, Multivariate Statistics.

Proposed new course. *Trustworthy AI for Cyber-Physical Security and Safety*. A graduate seminar on what guarantees a model must provide before its output can guide a physical action, combining geometric deep learning, uncertainty quantification, and state-space models of physical processes.

Appointments

Graduate Research Assistant, Cyber Threat Intelligence Lab

Aug. 2023 – Present

Louisiana State University, Baton Rouge, LA

Advisor: Dr. Elias Bou-Harb

- Study what a deep learning model must demonstrate before its output can guide a decision in a running cyber-physical system (CPS), and build systems that meet that standard. Industrial control systems (ICS) and vehicle networks are the main settings.
- Show that deep learning detectors in CPS are optimistically evaluated and lose their usefulness once deployed, raising false alarms so often that operators cannot act on them. Develop adaptive conformal calibration that brings the false alarm rate back to the level the detector was calibrated for, without retraining it.
- Build ICS detectors on the defender's knowledge of the plant. Topological neural networks encode how the system is organized (Purdue model), and symbolic regression learns its physics from normal operation as equations an engineer can check.

Visiting Researcher, Software and Systems Engineering Group

Feb. 2022 – Jun. 2022

CISUC, University of Coimbra, Portugal

- Translated random forest models into loop-free imperative programs for formal verification in safety-critical settings, published at IEEE/IFIP DSN-W 2025.

Research Presentations

• IEEE/IFIP NOMS 2026

Rome, Italy

Oral Presentation; *GRACE-ICS: Graph-Aware Conformal Intervals for Industrial Operations Assurance*

May 2026

• IEEE/IFIP NOMS 2025

Honolulu, HI, USA

Oral Presentation; *Enhancing Network Security Management in Water Systems using FM-based Attack Attribution*

May 2025

Schools & Symposia Attended

• 15th Annual Machine Learning Symposium

New York, NY, USA

Participant

Oct. 2024

• 9th PhD Summer School in Discrete Mathematics

Rogla, Slovenia

Participant

Jul. 2019

Professional Service

- Session Chair, Experience Session, IEEE/IFIP Network Operations and Management Symposium (NOMS), Rome, Italy, 2026.
- Reviewer, IEEE Transactions on Network and Service Management (TNSM).
- Student Member, IEEE Communications Society.

Grants & Scholarships

- **\$25,900** | University of Primorska Scholarship for Outstanding Achievements, Koper, Slovenia, 2017–2022.
- **\$2,700** | ERASMUS+ Thesis Research Scholarship, University of Coimbra, Portugal, 2022.

- **\$2,350** | ERASMUS+ Exchange Scholarship, Eötvös Loránd University, Budapest, Hungary, 2021.
- **\$1,950** | ERASMUS+ Traineeship Scholarship, Vienna, Austria, 2021.
- **\$500** | Student Travel Grant, IEEE/IFIP NOMS, Honolulu, HI, USA, 2025.

Amounts awarded in euros are shown in U.S. dollars at the September 2026 exchange rate (1 EUR = 1.15 USD)

Honors & Awards

- Gold Medal, Prmatijada Student Mathematical Competition, Golden Sands, Bulgaria, 2019.
- Competitor, Al-Khorezmi International Mathematical Olympiad, Urgench, Uzbekistan, 2018.

Industry Experience

Business Intelligence Consultant / Data Warehouse Engineer Aug. 2022 – Aug. 2023
CRMT, Ljubljana, Slovenia

- Built data warehouse and business intelligence solutions with SQL and ETL pipelines for clients in insurance, retail, and telecommunications.
- Developed applied machine learning models for public sector and telecommunications clients, including a recommender system deployed in a production ETL pipeline.

Data Analyst Trainee Jul. 2021 – Sep. 2021
ClearKarma, Vienna, Austria

- Supported market analysis, data extraction, and prototype development for data-driven product features.

Technical Skills

Programming. Python, Java, R, SQL, LaTeX.

Machine learning and verification. PyTorch, scikit-learn, TopoX (TopoNetX, TopoModelX), PySR, OpenJML, CUDA and GPU training.

Data and systems. Git, Linux, ETL pipelines, data warehousing.

Research practice. Reproducible experiment pipelines, benchmarking frameworks, open-source release.

Languages. Serbian (native), English and Slovenian (full professional proficiency).

Professional References

- Dr. Elias Bou-Harb, Associate Professor and Ph.D. Advisor, Division of Computer Science and Engineering, Louisiana State University
ebouharb@lsu.edu
- Dr. Ibrahim Baggili, Roger Richardson Professor and Division Chair, Division of Computer Science and Engineering, Louisiana State University
ibaggili@lsu.edu
- Dr. Joseph Khoury, Assistant Professor, Division of Computer Science and Engineering, Louisiana State University
joseph.khoury@lsu.edu
- Dr. Mustafa Hajij, Assistant Professor of Artificial Intelligence, University of San Francisco, and Co-Founding Engineer, Vinci4D.ai
mhajij@usfca.edu